



Failure to compensate or failure to reward? A decomposition of inequality of opportunity

Kristof Bosmans¹ · Bart Dormans² · Z. Emel Öztürk³

Received: 24 March 2023 / Accepted: 28 April 2025
© The Author(s) 2025

Abstract

We develop new ex ante and ex post liberal measures of inequality of opportunity that decompose into compensation and reward parts. Roughly, the compensation part measures the inequality between individuals with different circumstances (e.g., parental background), whereas the reward part measures the extent to which income differences reflect the market returns to the exercise of individual responsibility (e.g., work effort). As it turns out, the compensation components of the measures take the form of utilitarian measures of inequality of opportunity, thereby revealing a striking connection between the rival liberal and utilitarian approaches. We apply our measures and their decompositions to the net and gross income distributions of ten European countries.

Keywords Inequality of opportunity · Income inequality · Compensation · Reward · Decomposition

1 Introduction

In the past few decades, political philosophers and public economists have proposed various versions of equality of opportunity.¹ These proposals have in common that they condemn

¹ Key contributions by philosophers are Arneson (1989), Cohen (1989), Dworkin (1981a, b) and Rawls (1971, 1982). For overviews of the economic literature on equality of opportunity, see Ferreira and Peragine (2016), Fleurbaey (2008), Fleurbaey and Maniquet (2011), Ramos and Van de gaer (2016) and Roemer and Trannoy (2016).

✉ Kristof Bosmans
k.bosmans@maastrichtuniversity.nl
Bart Dormans
b.l.e.dormans@tilburguniversity.edu
Z. Emel Öztürk
z.e.ozturk@tilburguniversity.edu

¹ Department of Microeconomics and Public Economics, Maastricht University, Tongersestraat 53, 6211 LM Maastricht, The Netherlands

² Department of Strategy and Entrepreneurship, Tilburg University, 5000 LE Tilburg, The Netherlands

³ Department of Economics, Tilburg University, 5000 LE Tilburg, The Netherlands

inequality only insofar as it is caused by factors beyond the sphere of individual responsibility. If, for example, one individual is worse off than another due to a difference in parental background—a typical non-responsibility factor, or circumstance—then the inequality between these two individuals is morally problematic. But if the inequality is instead caused by a difference in work effort—a typical responsibility factor—then it is morally unproblematic.

More recently, the actual distributive justice judgments of participants in experiments and questionnaires have been shown to be aptly described by the moral ideal of equality of opportunity.² The societal relevance of the ideal is further attested to by the incorporation of criteria and measures of equality of opportunity into the policy evaluation apparatus by institutions such as the World Bank.³

The so-called liberal approach to equality of opportunity formalizes the ideal by use of two principles.⁴ The *principle of compensation* condemns the income differences caused by individual circumstances, requiring compensation to eliminate them. The *principle of reward* approves of the income differences that result from the exercise of responsibility, demanding that they be treated as just rewards. Note that the approach is referred to as “liberal” because its reward principle requires respect for the laissez-faire market returns to the exercise of responsibility, whereas the alternative “utilitarian” approach employs a reward principle that declares any rewards for the exercise of responsibility as equally good.⁵

Interestingly, the principles of compensation and reward will often offer sharply distinct evaluations of the very same policies or other phenomena that affect the income distribution. For example, policies that diminish the inequality of income—such as progressive taxation—are likely to be beneficial in terms of compensation, but detrimental in terms of reward. It is, moreover, possible that two income distributions exhibit similar levels of inequality of opportunity, while differing considerably in how they fare with respect to compensation and reward. For example, a progressive taxation policy and a non-intervention policy may have the same effect on inequality of opportunity: the former through its impact on compensation and the latter through its impact on reward. Therefore, to truly understand inequality of opportunity, and to design policies that properly address it, it is important to establish how much of the observed level of inequality of opportunity stems from a failure to compensate and how much from a failure to reward.

However, no attempt has been made so far to measure how well an income distribution performs in terms of compensation and reward separately, and to gauge the contribution of each to overall inequality of opportunity. The main objective of this study is to fill this gap in the literature.

Our first contribution is to develop new liberal inequality of opportunity measures that decompose as the sum of two parts, where one part represents compensation unfairness and the other part represents reward unfairness. Roughly speaking, the compensation component is a measure of the inequality among individuals with different circumstances, whereas the reward component is a measure of the extent to which the income differences between same-circumstance individuals deviate from the (morally desired) laissez-faire income differences. The newly developed liberal measures capture the so-called *ex ante* and *ex post* perspectives

² See, for example, Alesina and La Ferrara (2005), Almås et al. (2020), Cappelen et al. (2007), Cappelen et al. (2013), Fong (2001), Konow (2000) and Schokkaert and Devooght (2003).

³ An example of such a measure is the World Bank’s Human Opportunity Index. With its Centre for Opportunity and Equality, also the OECD shows a commitment to research on equality of opportunity.

⁴ Bossert (1995), Bossert and Fleurbaey (1996) and Fleurbaey (1994, 1995) were the first to define the two principles.

⁵ On the comparison of liberal and utilitarian reward, see in particular Fleurbaey (2008).

on inequality of opportunity. Our *ex ante* measure relies solely on circumstance information, whereas our *ex post* measure relies solely on responsibility information.

Although we build our analysis on the liberal approach, it turns out that a great deal can be said also about the utilitarian approach. Indeed, we find, strikingly, that utilitarian measures of inequality of opportunity emerge as the compensation components of our liberal measures. Therefore, we uncover an unexpected link between the liberal and utilitarian approaches—approaches that are usually viewed as profoundly distinct—that helps to unify and harmonize the literature.

Our second contribution is to apply our newly developed measures and their decompositions to ten European countries. In particular, we consider how income taxes affect inequality of opportunity. We find that the tax systems in the considered countries virtually all reduce the compensation component. The reward component, on the other hand, increases by construction. The effect of the tax system on overall inequality of opportunity therefore depends on the sum of two opposite effects. For several countries, the increasing effect of reward unfairness dominates the decreasing effect of compensation unfairness, resulting in the surprising finding that their tax systems increase overall inequality of opportunity.

The application also showcases how our new measures and decompositions render it possible to study the liberal and utilitarian approaches to inequality of opportunity measurement within a single unified framework. We believe this in itself to be a fruitful addition to the empirical analysis of inequality of opportunity.

In the next section, we introduce the notation. Section 3 develops our two new absolute measures of inequality of opportunity, and Section 4 decomposes these measures into compensation and reward parts. In Section 5, we repeat the analysis of the preceding sections for the case of relative, rather than absolute, measures of inequality of opportunity. Section 6 presents our empirical application. Section 7 concludes.

2 Notation

Society is a finite set N of n individuals. We let $\mathcal{N}$ denote the set of all societies. For a society N , we let $x_N = (x_i)_{i \in N}$ denote an income distribution, where the real number x_i is the income of individual i . The set of possible income distributions is $X = \bigcup_{N \in \mathcal{N}} \mathbb{R}^n$. The average income in x_N is $\bar{x}_N = \sum_{i \in N} x_i / n$.

Individuals differ in their circumstance and responsibility characteristics. Individual i 's circumstance and responsibility characteristics are denoted by c_i and r_i , respectively. The sets of possible circumstance and responsibility characteristics are C and R , respectively. For a society N , we let $c_N = (c_i)_{i \in N}$ and $r_N = (r_i)_{i \in N}$, and refer to the pair (c_N, r_N) in $C^n \times R^n$ as the characteristics profile. The set of possible characteristics profiles is $P = \bigcup_{N \in \mathcal{N}} (C^n \times R^n)$.

We use $N(c)$ to denote the subset of N that collects all individuals with circumstance characteristics equal to c , and refer to such a subset as a circumstance group. Similarly, we let $N(r)$ denote the subset of N that collects all individuals with responsibility characteristics equal to r , and refer to it as a responsibility group. The cardinalities of $N(c)$ and $N(r)$ are $n(c)$ and $n(r)$. We use $x_{N(c)} = (x_i)_{i \in N(c)}$ and $x_{N(r)} = (x_i)_{i \in N(r)}$ to denote the income distributions in $N(c)$ and $N(r)$, and let $\bar{x}_{N(c)} = \sum_{i \in N(c)} x_i / n(c)$ and $\bar{x}_{N(r)} = \sum_{i \in N(r)} x_i / n(r)$.

The reward principle in the liberal approach to equality of opportunity requires respecting the market returns to the exercise of responsibility (see Subsection 3.1). For this reason, we

single out the distribution of laissez-faire market incomes.⁶ For a society N , the market income distribution is $m_N = (m_i)_{i \in N}$ in X . We assume that there exists a function $m : C \times R \rightarrow \mathbb{R}$ that assigns to each combination of circumstance and responsibility characteristics (c, r) in $C \times R$ a market income $m(c, r)$.⁷ Hence, for each individual i in N , we have $m_i = m(c_i, r_i)$. Clearly, the market income function m specifies how the market rewards combinations of circumstance and responsibility characteristics.

3 Two new inequality of opportunity measures

We measure inequality of opportunity in an income distribution x_N in two steps. In the first step, we determine the fair income distribution—on the basis of the principles of equality of opportunity—of the total income in x_N . In the second step, we define inequality of opportunity as the extent to which x_N diverges from the fair income distribution. The key references for this two-step approach to inequality of opportunity measurement are Almås et al. (2011), Devooght (2008) and Fleurbaey and Schokkaert (2009).⁸

3.1 The first step: determining the fair income distribution

In the first step, we employ a rule that delivers the fair income distribution as an output using the relevant ethical information as an input. Formally, a fair division rule $F : X \times P \rightarrow X$ associates to each income distribution and each characteristics profile $(x_N; c_N, r_N)$ in $X \times P$ a fair income distribution x_N^F in X such that $\sum_{i \in N} x_i^F = \sum_{i \in N} x_i$.⁹ The fair division rules that we are interested in receive their ethical content from the two principles of equality of opportunity: compensation and reward.

The principle of compensation says that income differences due to circumstance characteristics ought to be eliminated. We consider two versions of the principle: the ex post version uses only information on responsibility characteristics, whereas the ex ante version uses only information on circumstance characteristics.¹⁰

Ex post compensation says that individuals with equal responsibility characteristics should receive the same income. The axiom was introduced by Bossert (1995) and Bossert and Fleurbaey (1996), and underlies the approach of Roemer (1993).¹¹

⁶ These are the hypothetical incomes that would result if government interference in the government were minimal. Various strands of libertarian or classical liberal thought disagree on what “minimal” interference entails (Mack and Gaus 2004). In our empirical application (Section 6), we follow Almås et al. (2011) and Devooght (2008) and do not estimate hypothetical laissez-faire income, but rather proxy it by gross income.

⁷ This assumption is standard in the literature on redistribution rules. See the references in Subsection 3.1.

⁸ See Hufe et al. (2022) and Mahler and Ramos (2019) for recent uses of the approach.

⁹ The fair division rules that we study here use very limited information about the current income distribution x_N : only the amount of total income. Of course, one can construct rules that use more information, e.g., conservative rules that prescribe preserving parts of x_N .

¹⁰ Bosmans and Öztürk (2021, 2022), Fleurbaey and Peragine (2013) and Ooghe et al. (2007) pay particular attention to the comparison between the ex post and ex ante perspectives. Note that in empirically implementing these compensation axioms, the informational basis may become blurred. For example, a common measure of responsibility—in particular, of work effort—is the income rank position of an individual in his circumstance group (Roemer 1993). This measure of responsibility would make the ex post compensation principle defined below depend on circumstance information. But this is a matter of practical implementation, not of meaning at the conceptual level.

¹¹ Bossert (1995) and Bossert and Fleurbaey (1996) also study various strengthenings and weakenings of the axiom.

Ex post compensation For each income distribution x_N in X , each characteristics profile (c_N, r_N) in P and all individuals i and j in N , if $r_i = r_j$, then $x_i^F = x_j^F$.

Ex ante compensation says that all circumstance groups should receive the same average income. The axiom was introduced by Checchi and Peragine (2010), Fleurbaey and Peragine (2013), and Ooghe et al. (2007), and underlies the approach of Van de gaer (1993).

Ex ante compensation For each income distribution x_N in X , each characteristics profile (c_N, r_N) in P and all individuals i and j in N , we have $\bar{x}_{N(c_i)}^F = \bar{x}_{N(c_j)}^F$.

The average income of circumstance group $N(c_i)$ can be interpreted as the value of the income opportunities available to individual i , that is, the incomes that i can attain by “choosing” a responsibility characteristic among those occurring in his circumstance group. In this interpretation, ex ante compensation calls for the equalization of the value of income opportunities across individuals.¹² Note that alternative versions of ex ante compensation exist. For example, Lefranc et al. (2008, 2009) propose to measure the value of an individual’s income opportunities by the Gini social welfare level associated with the income distribution of his circumstance group.

We now turn to the principle of liberal reward. This principle says that laissez-faire market income differences due to responsibility characteristics should be preserved.¹³ Again, we consider two versions of the principle, with the ex ante version only using information on circumstance characteristics and the ex post version only using information on responsibility characteristics.

Ex ante reward says that two individuals with the same circumstance characteristics—that is, individuals who differ (if at all) only in terms of responsibility—should be subject to the same transfer, where a transfer is defined as the difference between allocated income and market income. In other words, the income difference between these two individuals should be equal to the difference between their market incomes. The axiom was introduced by Bossert (1995) and Bossert and Fleurbaey (1996).

Ex ante reward For each income distribution x_N in X , for each characteristics profile (c_N, r_N) in P and for all individuals i and j in N , if $c_i = c_j$, then $x_i^F - m_i = x_j^F - m_j$.

Ex post reward says that all responsibility groups should be subject to the same average transfer, that is, the same difference between average allocated income and average market income. Equivalently, the average income difference between two responsibility groups should be equal to the difference between their average market incomes. This axiom was introduced by Trannoy (2017).

Ex post reward For each income distribution x_N in X , for each characteristics profile (c_N, r_N) in P and for all individuals i and j in N , we have $\bar{x}_{N(r_i)}^F - \bar{m}_{N(r_i)} = \bar{x}_{N(r_j)}^F - \bar{m}_{N(r_j)}$.

Note that with these two reward axioms, the literature has put the stress on absolute inequality. Indeed, these axioms require respecting absolute differences between market

¹² See Ooghe et al. (2007) on this interpretation.

¹³ Note that the principle can capture alternative (non-liberal) ideals of reward by letting m represent, instead of market income, an income standard that reflects how the exercise of responsibility ought to be rewarded according to some external theory of desert. See Fleurbaey (2008, pp. 221–223) and Roemer (2012, pp. 178–179).

incomes. Accordingly, we will incorporate these axioms into absolute measures of inequality of opportunity in the rest of this section. We consider more recently proposed relative versions of reward and the corresponding relative inequality of opportunity measures in Section 5.

Table 1 illustrates the four axioms for society $N = \{1, 2, 3, 4, 5\}$, the members of which differ in terms of circumstance (talent) and responsibility (effort) characteristics. Total income is 340 in m_N and 400 in the other eight income distributions. The equal income distribution x_N^1 is consistent with both compensation axioms. The laissez-faire income distribution x_N^2 , which preserves the income differences present in the market income distribution m_N (as each individual receives what he has in m_N plus 12), is consistent with both reward axioms. The income distributions x_N^3 to x_N^6 each are consistent with only one out of the four axioms.¹⁴

Two fair division rules emerge from the four axioms and will be the basis for our inequality of opportunity measures. The ex ante rule is the only rule to combine ex ante compensation and ex ante reward (Bossert et al. 1999; Bosmans and Öztürk 2022), whereas the ex post rule is the only rule to combine ex post compensation and ex post reward (Bosmans and Öztürk 2022).¹⁵ Note that various alternative fair division rules have been studied in the literature (e.g., Bossert 1995, Bossert and Fleurbaey 1996, Cappelen and Tungodden 2009, Cappelen and Tungodden 2017 and Iturbe-Ormaetxe 1997).

The ex ante rule gives each individual the difference between his market income and the average market income of his circumstance group, plus a constant to ensure that the total income in x_N is fully distributed.¹⁶

Ex ante rule For each income distribution x_N in X , each characteristics profile (c_N, r_N) in P and each individual i in N , we have

$$x_i^{EA} = m_i - \bar{m}_{N(c_i)} + \bar{x}_N. \quad (1)$$

The ex post rule gives each individual the average market income of his responsibility group, plus a constant to ensure that the total income in x_N is fully distributed.

Ex post rule For each income distribution x_N in X , each characteristics profile (c_N, r_N) in P and each individual i in N , we have

$$x_i^{EP} = \bar{m}_{N(r_i)} + \bar{x}_N - \bar{m}_N. \quad (2)$$

¹⁴ In x_N^3 , income is equally distributed within each responsibility group, as demanded by ex post compensation. In x_N^4 , both circumstance groups receive the same average income (equal to 80), as required by ex ante compensation. In x_N^5 , each member of the high-talent circumstance group receives the same transfer (equal to 9) and so does each member of the low-talent circumstance group (transfer equal to 14), as required by ex ante reward. In x_N^6 , each responsibility group receives the same average transfer (equal to 12), as demanded by ex post reward.

¹⁵ Bossert et al. (1999) refer to the ex ante rule as the “observable average conditionally egalitarian” rule, and to the ex post rule as the “observable average egalitarian equivalent” rule.

¹⁶ Bosmans and Öztürk (2022) characterize the ex ante rule under the assumption that total income in x equals total income in m . However, any surplus ($\sum_{i \in N} x_i - \sum_{i \in N} m_i > 0$) or shortfall ($\sum_{i \in N} x_i - \sum_{i \in N} m_i < 0$) must be divided equally, as in Eq. 1. Indeed, ex ante compensation implies that each circumstance group should receive the same part (in per capita terms) of the surplus or shortfall, and ex ante reward implies that this part should be shared equally among all members of a circumstance group. The same reasoning applies to the ex post rule in Eq. 2.

Table 1 An example for a society of five individuals

Individual	Talent	Effort	m_N	x_N^1	x_N^2	x_N^3	x_N^4	x_N^5	x_N^6	x_N^{EA}	x_N^{EP}
1	high	high	120	80	132	106	89	129	70	105	92
2	high	low	70	80	82	41	71	79	44	55	62
3	low	high	60	80	72	106	80	74	110	90	92
4	low	high	60	80	72	106	70	74	96	90	92
5	low	low	30	80	42	41	90	44	80	60	62
Ex post compensation			✗	✓	✗	✓	✗	✗	✗	✗	✓
Ex ante compensation			✗	✓	✗	✗	✓	✗	✗	✓	✗
Ex ante reward			✓	✗	✓	✗	✗	✓	✗	✓	✗
Ex post reward			✓	✗	✓	✗	✗	✗	✓	✗	✓

The final two columns of Table 1 show the income distributions chosen by the ex ante and ex post rules for a total income of 400.

As mentioned before, we measure the inequality of opportunity in an income distribution x_N by the extent to which it diverges from the fair income distribution as chosen by the ex ante or ex post rule. Therefore, we now turn to the measurement of divergence.

3.2 The second step: measuring divergence

A divergence measure is a function $D : \bigcup_{N \in \mathcal{N}} (\mathbb{R}^n \times \mathbb{R}^n) \rightarrow \mathbb{R}$ that assigns to each pair of income distributions x_N and y_N a divergence level $D(x_N, y_N)$. In particular, we use the divergence measure proposed by Cowell (1985, Corollary to Theorem 3):¹⁷ for all income distributions x_N and y_N in X ,

$$D_\alpha(x_N, y_N) = \begin{cases} \frac{1}{n} \sum_{i \in N} \{\exp\{\alpha[x_i - y_i - (\bar{x}_N - \bar{y}_N)]\} - 1\} & \text{if } \alpha \neq 0, \\ \frac{1}{n} \sum_{i \in N} [x_i - y_i - (\bar{x}_N - \bar{y}_N)]^2 & \text{if } \alpha = 0. \end{cases} \quad (3)$$

We stress three properties of the divergence measure D_α that are useful in what follows. First, the measure D_α is absolute, i.e., adding the same amount to each individual in either of the income distributions does not impact divergence.¹⁸ Second, the measure D_α is decomposable, i.e., for any partitioning of the population into subgroups, divergence can be written as the sum of the divergence within the subgroups and the divergence between the subgroups.¹⁹

¹⁷ Other studies have used alternative divergence measures to measure inequality of opportunity. Hufe et al. (2022) and Mahler and Ramos (2019) use the relative divergence measures of Magdalou and Nock (2011), Devooght (2008) uses the relative version of Cowell's (1985) divergence measure (which we also use in Section 5) and Almås et al. (2011) develop their own divergence measure. We use Cowell's (1985) divergence measure in Eq. 3 because it is absolute and decomposable, two properties (defined below) that are essential for our particular purpose. The other divergence measures violate at least one of these two properties.

¹⁸ Formally, for all income distributions x_N and y_N in X and all real numbers κ and λ , we have $D_\alpha(x_N, y_N) = D_\alpha((x_i + \kappa)_{i \in N}, (y_i + \lambda)_{i \in N})$.

¹⁹ See Supplementary Material for the formal definition of decomposability.

Third, the divergence measure D_α simplifies to an income inequality measure $I : X \rightarrow \mathbb{R}$ if y_N is an equal distribution. That is, for all x_N and y_N in X such that $y_i = \bar{y}_N$ for each i in N , we have that $D_\alpha(x_N, y_N)$ equals²⁰

$$I_\alpha(x_N) = \begin{cases} \frac{1}{n} \sum_{i \in N} \{\exp[\alpha(x_i - \bar{x}_N)] - 1\} & \text{if } \alpha \neq 0, \\ \frac{1}{n} \sum_{i \in N} (x_i - \bar{x}_N)^2 & \text{if } \alpha = 0. \end{cases} \quad (4)$$

For $\alpha = 0$, the measure I_α is the variance. For $\alpha < 0$, it is ordinally equivalent to the Kolm (1976) family of income inequality measures. The measure I_α inherits absoluteness and decomposability from the measure D_α .²¹

3.3 Two inequality of opportunity measures

An inequality of opportunity measure is a function $J : X \times P \rightarrow \mathbb{R}$ that assigns to each income distribution and corresponding characteristics profile $(x_N; c_N, r_N)$ in $X \times P$ a level of inequality of opportunity $J(x_N; c_N, r_N)$. In our notation, we suppress the dependence of J on (c_N, r_N) and simply use $J(x_N)$ to denote the level of inequality of opportunity.

In accordance with the two steps described above, we measure inequality of opportunity in an income distribution x_N by the extent to which it diverges from the fair income distribution x_N^F . That is, for each income distribution x_N in X ,

$$J_\alpha^F(x_N) = D_\alpha(x_N, x_N^F). \quad (5)$$

The inequality of opportunity measure in Eq. 5 is absolute.²² We discuss the relative counterpart of the measure in Section 5.

By applying Eq. 5 to the ex ante and ex post rules, we obtain two distinct inequality of opportunity measures. Note that the constants in Eqs. 1 and 2 drop out because the inequality of opportunity measure is absolute.

Ex ante inequality of opportunity measure For each income distribution x_N in X ,

$$J_\alpha^{EA}(x_N) = D_\alpha(x_N, (m_i - \bar{m}_{N(c_i)})_{i \in N}). \quad (6)$$

Ex post inequality of opportunity measure For each income distribution x_N in X ,

$$J_\alpha^{EP}(x_N) = D_\alpha(x_N, (\bar{m}_{N(r_i)})_{i \in N}). \quad (7)$$

Table 2 presents the ex ante and ex post inequality of opportunity levels (for $\alpha = 0$) for the income distributions in Table 1.

²⁰ On the inequality measure I_α , see Bosmans and Cowell (2010) and Chakravarty and Tyagarupananda (1998, 2009). The relationship between the measures D_α and I_α is simple: we have $D_\alpha(x_N, y_N) = I_\alpha(x_N - y_N)$ for all x_N and y_N in X .

²¹ An income inequality measure I is absolute if, for each income distribution x_N in X and each real number κ , we have $I(x_N) = I((x_i + \kappa)_{i \in N})$. For a formal definition of decomposability, see Supplementary Material.

²² Formally, for each income distribution x_N in X and each real number κ , we have $J_\alpha^F(x_N) = J_\alpha^F((x_i + \kappa)_{i \in N})$.

Table 2 Inequality of opportunity measures and their decompositions

Individual	Talent	Effort	m_N	x_N^1	x_N^2	x_N^3	x_N^4	x_N^5	x_N^6	x_N^{EA}	x_N^{EP}
1	high	high	120	80	132	106	89	129	70	105	92
2	high	low	70	80	82	41	71	79	44	55	62
3	low	high	60	80	72	106	80	74	110	90	92
4	low	high	60	80	72	106	70	74	96	90	92
5	low	low	30	80	42	41	90	44	80	60	62
	Ex post compensation		X	✓	X	✓	X	X	X	X	✓
	Ex ante compensation		X	✓	X	X	✓	X	X	✓	X
	Ex ante reward		✓	X	✓	X	X	✓	X	✓	X
	Ex post reward		✓	X	✓	X	X	X	✓	X	✓
	J_0^{EA}			370	486	214	382.4	384	436.4	0	46
	J_0^{EAC}			0	486	28.2	0	384	352.7	0	6
	J_0^{EAR}			370	0	185.8	382.4	0	83.7	0	40
	J_0^{EP}			216	640	294	300.4	526	294.4	46	0
	J_0^{EPC}			0	640	0	72.2	525.8	294.4	32.5	0
	J_0^{EPR}			216	0	294	228.2	0.2	0	13.5	0
	I_0			0	856	1014	72.4	754	510.4	370	216

4 Decomposition into compensation and reward parts

Table 2 illustrates that income distributions may exhibit similar levels of inequality of opportunity, while differing strongly in terms of their performance with respect to compensation and reward. For example, income distributions x_N^4 and x_N^5 exhibit similar inequality of opportunity as measured by J_0^{EA} , but x_N^4 exhibits only (ex ante) reward unfairness, whereas x_N^5 exhibits only (ex ante) compensation unfairness. Likewise, income distributions x_N^3 and x_N^6 are similar in terms of inequality of opportunity as measured by J_0^{EP} , but x_N^3 exhibits only (ex post) reward unfairness, whereas x_N^6 exhibits only (ex post) compensation unfairness. To truly understand what drives inequality of opportunity, one needs to be able to measure what part of it is due to each of the two sources of unfairness.

Fortunately, the ex ante and ex post measures of inequality of opportunity turn out to decompose neatly into compensation and reward parts. We discuss the decompositions here and provide their derivations in Supplementary Material.

First, consider the decomposition of the ex ante inequality of opportunity measure. We use $C(N)$ to denote the set of circumstance characteristics observed in N .

Decomposition of the ex ante inequality of opportunity measure For each income distribution x_N in X ,

$$J_\alpha^{EA}(x_N) = J_\alpha^{EAC}(x_N) + J_\alpha^{EAR}(x_N), \quad (8)$$

where the compensation component is

$$J_\alpha^{EAC}(x_N) = I_\alpha((\bar{x}_{N(c_i)})_{i \in N}), \quad (9)$$

and the reward component is

$$J_\alpha^{EAR}(x_N) = \sum_{c \in C(N)} w_{\alpha c} I_\alpha((x_i - m_i)_{i \in N(c)}) \quad \text{with} \quad w_{\alpha c} = \frac{n(c)}{n} \frac{\exp(\alpha \bar{x}_{N(c)})}{\exp(\alpha \bar{x}_N)}. \quad (10)$$

The two components J_{α}^{EAC} and J_{α}^{EAR} neatly capture the intuitions of the corresponding compensation and reward axioms. The compensation component J_{α}^{EAC} measures the inequality of average income between circumstance groups, which the ex ante compensation axiom condemns. The reward component J_{α}^{EAR} measures the inequality of transfers within circumstance groups (as a weighted sum of the circumstance group inequality levels), disapproved of by the ex ante reward axiom. Note that the weights in J_{α}^{EAR} are population weights if $\alpha = 0$, i.e., if I_{α} is the variance.

Next, we turn to the decomposition of the ex post inequality of opportunity measure. We use $R(N)$ to denote the set of responsibility characteristics observed in N .

Decomposition of the ex post inequality of opportunity measure For each income distribution x_N in X ,

$$J_{\alpha}^{EP}(x_N) = J_{\alpha}^{EPC}(x_N) + J_{\alpha}^{EPR}(x_N), \quad (11)$$

where the compensation component is

$$J_{\alpha}^{EPC}(x_N) = \sum_{r \in R(N)} w_{\alpha r} I_{\alpha}(x_{N(r)}) \quad \text{with} \quad w_{\alpha r} = \frac{n(r) \exp[\alpha(\bar{x}_{N(r)} - \bar{m}_{N(r)})]}{n \exp[\alpha(\bar{x}_N - \bar{m}_N)]}, \quad (12)$$

and the reward component is

$$J_{\alpha}^{EPR}(x_N) = I_{\alpha}((\bar{x}_{N(r_i)} - \bar{m}_{N(r_i)})_{i \in N}). \quad (13)$$

Again, the two components J_{α}^{EPC} and J_{α}^{EPR} capture the intuitions of the corresponding compensation and reward axioms well. The compensation component J_{α}^{EPC} measures the inequality within responsibility groups (as a weighted sum of responsibility group inequality levels), condemned by the ex post compensation axiom. Note that the weights in J_{α}^{EPC} are population weights if $\alpha = 0$, i.e., if I_{α} is the variance. The reward component J_{α}^{EPR} measures the inequality of average transfers between responsibility groups, which the ex post reward axiom disapproves of.

Table 2 presents the decompositions of overall inequality of opportunity into compensation and reward parts for our six income distributions. Note that if an income distribution is consistent with a particular axiom, then the corresponding component is zero.

We end this section with a discussion of how the decomposition helps to connect the two key approaches in the literature on inequality of opportunity measurement: the liberal and utilitarian approaches. Our ex ante and ex post measures belong to the liberal approach as they are based on the liberal versions of the reward principle defined in Subsection 3.1. The utilitarian approach is based on utilitarian versions of reward that call for distributional neutrality with regard to rewarding the exercise of responsibility.²³

Now, an important observation is that the measures J_{α}^{EAC} and J_{α}^{EPC} are in the form, respectively, of utilitarian ex ante and ex post measures.²⁴ Our decomposition thus reveals a striking connection between the liberal and utilitarian approaches: the ex ante utilitarian measure J_{α}^{EAC} emerges as the compensation component of the ex ante liberal measure J_{α}^{EA} , and, similarly, the ex post utilitarian measure J_{α}^{EPC} emerges as the compensation

²³ The measures are referred to as “utilitarian” because distributional neutrality is a key property of utilitarianism.

²⁴ In an influential contribution, Checchi and Peragine (2010) have proposed procedures to derive utilitarian inequality of opportunity measures from income inequality measures. Applying their ex ante procedure to I_{α} delivers exactly J_{α}^{EAC} . Applying their ex post procedure to I_{α} delivers a version of J_{α}^{EPC} in which the weights are replaced by population weights. So their ex post utilitarian measure for I_{α} coincides with J_{α}^{EPC} only if $\alpha = 0$.

component of the ex post liberal measure J_{α}^{EP} . This means that the liberal and utilitarian families of inequality of opportunity measures, which are usually treated as belonging to profoundly distinct approaches, are actually closely connected. It moreover means that the empirical results yielded by the oft-applied utilitarian approach are of immediate relevance also to researchers mainly interested in applying the liberal approach.²⁵ Indeed, the utilitarian measures each serve as a lower bound for the corresponding liberal measure.

This connection between the liberal and utilitarian approaches also helps to illuminate the relationship between inequality of opportunity and income inequality. Consider first the ex ante case. Decomposing the income inequality measure I_{α} by circumstance groups yields, for each income distribution x_N in X ,

$$I_{\alpha}(x_N) = J_{\alpha}^{EAC}(x_N) + \sum_{c \in C(N)} v_{\alpha c} I_{\alpha}(x_{N(c)}), \quad (14)$$

where

$$v_{\alpha c} = \frac{n(c)}{n} \frac{\exp(\alpha \bar{x}_{N(c)})}{\exp(\alpha \bar{x}_N)}.$$

Clearly, the ex ante utilitarian measure J_{α}^{EAC} always takes a value that is less than or equal to that of the income inequality measure I_{α} . This relationship has given rise to an understanding of inequality of opportunity never being greater than income inequality, and correspondingly has often led practitioners to express inequality of opportunity as a share of income inequality. However, if one uses the liberal measure J_{α}^{EA} instead, then this understanding is no longer valid: the measure J_{α}^{EA} may very well take a greater value than does the income inequality measure I_{α} .²⁶ The reason is that the measure J_{α}^{EA} is the sum of J_{α}^{EAC} and J_{α}^{EAR} , where the value of J_{α}^{EAR} may exceed the value of the second term of Eq. 14.

Also in the ex post case, inequality of opportunity as measured by the liberal measure J_{α}^{EP} may exceed income inequality. Note, however, that decomposing I_{α} by responsibility groups yields J_{α}^{EPC} as a within component only if $\alpha = 0$.

5 The relative case

We now develop relative counterparts of the absolute inequality of opportunity measures discussed above, and provide their decompositions. In this section, we assume that incomes—including market incomes—are positive, i.e., we let $X = \bigcup_{N \in \mathcal{N}} \mathbb{R}_{++}^n$.²⁷

Following Bosmans and Öztürk (2022), we define relative versions of ex ante and ex post reward. These versions are defined in terms of relative transfers (ratio of allocated income to market income) instead of absolute transfers (difference between allocated and

²⁵ Note that especially the relative ex ante utilitarian measure has often been applied empirically (e.g., Belhaj Hassine 2012; Bourguignon et al. 2007; Brunori et al. 2019; Ferreira and Gignoux 2011; Hufe et al. 2017 and Marrero and Rodríguez 2013). In the next section, it becomes clear that the relationship discussed here also holds for the relative case.

²⁶ For the equal income distribution x_N^1 in Table 2, income inequality (I_0) and utilitarian inequality of opportunity (J_0^{EA}) are both equal to zero, whereas liberal inequality of opportunity (J_0^{EP}) is positive. The income distribution x_N^4 is another example where liberal inequality of opportunity (J_0^{EP}) exceeds income inequality (I_0).

²⁷ Otherwise we cannot exclude division by zero.

market income).²⁸ For ex ante and ex post compensation, the absolute and relative versions coincide.

Relative ex ante reward says that individuals with the same circumstance characteristics should be subject to the same relative transfer.

Relative ex ante reward For each income distribution x_N in X , each characteristics profile (c_N, r_N) in P and all individuals i and j in N , if $c_i = c_j$, then $x_i^F/m_i = x_j^F/m_j$.

Relative ex post reward says that all responsibility groups should be subject to the same relative average transfer.

Relative ex post reward For each income distribution x_N in X , each characteristics profile (c_N, r_N) in P and all individuals i and j in N , we have $\bar{x}_{N(r_i)}^F/\bar{m}_{N(r_i)} = \bar{x}_{N(r_j)}^F/\bar{m}_{N(r_j)}$.

Again, two fair division rules emerge from the axioms. The relative ex ante rule is the only rule to combine ex ante compensation and relative ex ante reward, whereas the relative ex post rule is the only rule to combine ex post compensation and relative ex post reward (Bosmans and Öztürk 2022).

Relative ex ante rule For each income distribution x_N in X , for each characteristics profile (c_N, r_N) in P and for each individual i in N , we have

$$x_i^{REA} = \frac{m_i}{\bar{m}_{N(c_i)}} \bar{x}_N. \quad (15)$$

Relative ex post rule For each income distribution x_N in X , for each characteristics profile (c_N, r_N) in P and for each individual i in N , we have

$$x_i^{REP} = \bar{m}_{N(r_i)} \frac{\bar{x}_N}{\bar{m}_N}. \quad (16)$$

To derive inequality of opportunity measures from the ex ante and ex post rules, we use the relative divergence measure D_ρ proposed by Cowell (1980a, 1985): for all income distributions x_N and y_N in X ,

$$D_\rho(x_N, y_N) = \begin{cases} \frac{1}{n\rho(\rho-1)} \sum_{i=1}^n \left[\left(\frac{x_i}{\bar{x}_N} \right)^\rho \left(\frac{y_i}{\bar{y}_N} \right)^{1-\rho} - 1 \right] & \text{for } \rho \neq 0, 1, \\ \frac{1}{n} \sum_{i=1}^n \frac{y_i}{\bar{y}_N} \ln \left(\frac{\bar{x}_N}{x_i} \frac{y_i}{\bar{y}_N} \right) & \text{for } \rho = 0, \\ \frac{1}{n} \sum_{i=1}^n \frac{x_i}{\bar{x}_N} \ln \left(\frac{x_i}{\bar{x}_N} \frac{\bar{y}_N}{y_i} \right) & \text{for } \rho = 1. \end{cases} \quad (17)$$

We stress three properties of the divergence measure D_ρ . First, the measure D_ρ is relative, i.e., multiplying each individual's income by the same factor in either of the income distributions does not impact divergence.²⁹ Second, the measure D_ρ is decomposable.³⁰ Third,

²⁸ The ex ante and ex post reward axioms in Subsection 3.1 are defined in terms of absolute transfers, which makes them less appropriate for relative inequality of opportunity measurement. Indeed, for two individuals i and j with equal transfers $x_i - m_i = x_j - m_j$, adding the same amount to x_i and x_j preserves the equality of the transfers, whereas multiplying x_i and x_j by the same factor typically does not.

²⁹ Formally, for all income distributions x_N and y_N in X and all positive real numbers κ and λ , we have $D_\rho(x_N, y_N) = D_\rho(\kappa x_N, \lambda y_N)$.

³⁰ See Supplementary Material for a formal definition.

D_ρ simplifies to the generalized entropy income inequality measure I_ρ if y_N is an equal distribution. That is, for all x_N and y_N in X such that $y_i = \bar{y}_N$ for each i in N , we have that $D_\alpha(x_N, y_N)$ equals³¹

$$I_\rho(x_N) = \begin{cases} \frac{1}{n\rho(\rho-1)} \sum_{i=1}^n \left[\left(\frac{x_i}{\bar{x}_N} \right)^\rho - 1 \right] & \text{for } \rho \neq 0, 1, \\ \frac{1}{n} \sum_{i=1}^n \ln \left(\frac{\bar{x}_N}{x_i} \right) & \text{for } \rho = 0, \\ \frac{1}{n} \sum_{i=1}^n \frac{x_i}{\bar{x}_N} \ln \left(\frac{x_i}{\bar{x}_N} \right) & \text{for } \rho = 1. \end{cases} \quad (18)$$

Setting $\rho = 0$ yields the mean logarithmic deviation and setting $\rho = 1$ delivers the Theil measure. For $\rho < 1$, the measure I_ρ is ordinally equivalent to the family of income inequality measures proposed by Atkinson (1970). The measure I_ρ inherits relativeness and decomposability from D_ρ .³²

To measure relative inequality of opportunity, we let, for each income distribution x_N in X ,

$$J_\rho^F(x_N) = D_\rho(x_N, x_N^F). \quad (19)$$

By applying Eq. 19 to the relative ex ante and ex post rules, we obtain two relative inequality of opportunity measures.³³

Relative ex ante inequality of opportunity measure For each income distribution x_N in X ,

$$J_\rho^{REA}(x_N) = D_\rho(x_N, (m_i/\bar{m}_{N(c_i)})_{i \in N}). \quad (20)$$

Relative ex post inequality of opportunity measure For each income distribution x_N in X ,

$$J_\rho^{REP}(x_N) = D_\rho(x_N, (\bar{m}_{N(r_i)})_{i \in N}). \quad (21)$$

Again, these two inequality of opportunity measures can be decomposed into compensation and reward parts. Note that the weights in J_ρ^{REAR} and in J_ρ^{REPC} are income shares if $\rho = 1$ and the weights in J_ρ^{REAR} are population weights if $\rho = 0$. We refer to Supplementary Material for the derivations.

Decomposition of the relative ex ante inequality of opportunity measure For each income distribution x_N in X ,

$$J_\rho^{REA}(x_N) = J_\rho^{REAC}(x_N) + J_\rho^{REAR}(x_N), \quad (22)$$

where the compensation component is

$$J_\rho^{REAC}(x_N) = I_\rho((\bar{x}_{N(c_i)})_{i \in N}), \quad (23)$$

³¹ On this family of income inequality measures, see Bourguignon (1979), Cowell (1980b), Cowell and Kuga (1981b, a) and Shorrocks (1980, 1984).

³² An income inequality measure I is relative if, for each income distribution x_N in X and each positive real number κ , we have $I(x_N) = I(\kappa x_N)$. See Supplementary Material for a definition of decomposability.

³³ Indeed, for each income distribution x_N in X and each positive real number κ , we have $J_\rho^F(x_N) = J_\rho^F(\kappa x_N)$.

and the reward component is

$$J_{\rho}^{REAR}(x_N) = \sum_{c \in C(N)} w_{\rho c} D_{\rho}(x_{N(c)}, m_{N(c)}) \quad \text{with} \quad w_{\rho c} = \frac{n(c)}{n} \frac{\bar{x}_{N(c)}^{\rho}}{\bar{x}_N^{\rho}}. \quad (24)$$

Decomposition of the relative ex post inequality of opportunity measure For each income distribution x_N in X ,

$$J_{\rho}^{REP}(x_N) = J_{\rho}^{REPC}(x_N) + J_{\rho}^{REPR}(x_N), \quad (25)$$

where the compensation component is

$$J_{\rho}^{REPC}(x_N) = \sum_{r \in R(N)} w_{\rho r} I_{\rho}(x_{N(r)}) \quad \text{with} \quad w_{\rho r} = \frac{n(r)}{n} \frac{\bar{x}_{N(r)}^{\rho}}{\bar{x}_N^{\rho}} \frac{\bar{m}_{N(r)}^{1-\rho}}{\bar{m}_N^{1-\rho}}, \quad (26)$$

and the reward component is

$$J_{\rho}^{REPR}(x_N) = D_{\rho}((\bar{x}_{N(r_i)})_{i \in N}, (\bar{m}_{N(r_i)})_{i \in N}). \quad (27)$$

The interpretations of these components are analogous to the interpretations of those in the absolute case. The only exception is that the relative ex ante and ex post reward components do not reduce further to simple measures of transfer inequality as do their absolute counterparts.

Also the points raised in the previous section about the relationship between the liberal and utilitarian inequality of opportunity measures and income inequality measures carry over from the absolute case. First, J_{ρ}^{REAC} and J_{ρ}^{REPC} are relative utilitarian ex ante and ex post measures. The version of J_{ρ}^{REAC} that uses the mean logarithmic deviation ($\rho = 0$) as an income inequality measure is particularly popular in empirical applications. Second, the measures J_{ρ}^{REA} and J_{ρ}^{REP} may take values greater than does the corresponding income inequality measure I_{ρ} .

6 Empirical application

We apply the ex ante and ex post measures of inequality of opportunity and their decompositions to ten European countries: Austria (AT), Belgium (BE), Bulgaria (BG), Germany (DE), Hungary (HU), Ireland (IE), Luxembourg (LU), the Netherlands (NL), Slovenia (SI) and the United Kingdom (UK). For each country, we measure inequality of opportunity in both the net and gross income distribution, allowing us to gauge the impact of taxes and benefits on inequality of opportunity. We apply both our absolute and relative measures.

6.1 Data description

We use the 2011 wave of the European Union - Statistics on Income and Living Conditions (EU-SILC) dataset.³⁴ The 2011 wave is of particular interest for our application, as it includes information on family background that we use as part of an individual's circumstance characteristics.

³⁴ Álvarez and Menéndez (2018a), Checchi and Peragine (2010) and Marrero and Rodríguez (2012) also use this dataset in their studies on inequality of opportunity.

We use observed individual incomes as our income variable.³⁵ Note that because the *ex ante* and *ex post* rules use observed incomes rather than incomes corresponding to hypothetical individual characteristics, we do not need to assume a specific functional form that maps circumstance and responsibility characteristics into income.³⁶ We use gross income to proxy for market income following Devooght (2008) and Almås et al. (2011). Gross income excluding all taxes is reported for all ten countries in our analysis, whereas net income is reported for all except for the Netherlands, Germany, the United Kingdom and Hungary. For these four countries, we have computed net income from gross income by using the then-applicable tax schemes.³⁷ To allow for comparisons between countries, we express all gross and net incomes in purchasing power parity (PPP) euros.³⁸ We winsorized gross and net incomes at the 99% level to reduce the possibly spurious impact of outliers.

As is common in the literature (e.g., Álvarez and Menéndez 2018b and Checchi and Peragine 2010), we impose a number of restrictions on our sample. We restrict the sample to (i) individuals working full-time or part-time, (ii) individuals aged between 25 and 60, (iii) individuals whose gross income is greater than zero, (iv) individuals for whom information about the education level of both parents is available, and (v) individuals whose father's main occupation is known and not in the armed forces occupations category. Restrictions (i) to (iii) limit the probability of alternative income means such as retirement and unemployment benefits. Restrictions (iv) and (v) allow for the use of parental characteristics as circumstances.

We use circumstance and responsibility variables that are common in the literature (see, e.g., Ferreira and Peragine 2016, Ramos and Van de gaer 2021 and Roemer and Trannoy 2016). Table 3 presents descriptive statistics.

We first explain how we selected our circumstance variables. We start with a set of potentially relevant variables: sex (male or female), immigration status (born in the same country as residence, or not), age (younger than 40, or older), whether the individual lived with both parents growing up (yes or no), the parental level of education (at least one of the parents medium or highly educated, or not), and the occupation of the father (highly-skilled occupation, or not).³⁹ These variables yield a baseline of 64 circumstance groups.

³⁵ We use individual incomes rather than equivalized household incomes as we are interested in how the circumstance and responsibility characteristics of individual household members translate into income. Equivalized income erases intra-household inequality, and is therefore not a suitable concept for our purposes. Almås et al. (2011), Checchi et al. (2016, p.10) and Devooght (2008) also use individual incomes rather than equivalized income.

³⁶ In contrast, the inequality of opportunity measures of Almås et al. (2011) and Devooght (2008) rely on incomes of hypothetical individuals, e.g., individuals with observed circumstance characteristics, but counterfactual responsibility characteristics that serve as a reference. Their parametric method thus necessitates the estimation of the market income function, whereas our non-parametric method does not. For a comparison of parametric and non-parametric methods in inequality of opportunity measurement, see Ramos and Van de gaer (2021).

³⁷ In fact, we have selected the ten countries because they are the only ones that report gross income excluding all taxes and that either report net income or allow for reliable computation of net income on the basis of the tax scheme of the country.

³⁸ As this adjustment entails multiplying each income in a country by the same conversion factor, it impacts the results of the absolute measures, but not those of the relative measures.

³⁹ In accordance with the International Standard Classification of Occupations (ISCO-08), high-skilled occupations are “managers”, “professionals”, “technicians and associated professionals”, and “clerical support workers”. We do not include the occupation of the mother as a potentially relevant circumstance variable because this variable is often missing in the dataset.

Table 3 Descriptive statistics (proportion of sample population)

	AT	BE	BG	DE	HU	IE	LU	NL	SI	UK
<i>Circumstance</i>										
Female	0.486	0.502	0.504	0.496	0.511	0.575	0.452	0.502	0.519	0.533
Immigrant	0.145	0.118	0.005	0.052	0.011	0.208	0.508	0.044	0.103	0.110
Aged 40-60	0.604	0.522	0.602	0.666	0.568	0.511	0.491	0.582	0.510	0.626
Both parents present	0.920	0.921	1.000	0.915	0.914	0.952	0.914	0.955	0.941	0.895
At least one parent medium or highly educated	0.676	0.578	0.574	0.922	0.468	0.631	0.522	0.674	0.390	0.499
Father's occupation: skilled	0.243	0.474	0.197	0.432	0.200	0.371	0.377	0.523	0.282	0.414
<i>Responsibility</i>										
Married	0.593	0.567	0.655	0.655	0.610	0.634	0.648	0.574	0.588	0.627
Hours worked										
Less than 20 hours	0.044	0.064	0.001	0.082	0.003	0.113	0.032	0.091	0.001	0.091
20-30 hours	0.205	0.159	0.027	0.152	0.053	0.193	0.162	0.240	0.048	0.152
More than 30 hours	0.751	0.777	0.972	0.765	0.944	0.694	0.806	0.670	0.952	0.757
Own education										
Upper secondary and lower	0.102	0.133	0.120	0.035	0.108	0.154	0.310	0.121	0.103	0.090
Post-secondary non-tertiary	0.660	0.354	0.594	0.514	0.623	0.251	0.364	0.418	0.557	0.435
First and second stage of tertiary	0.239	0.513	0.286	0.451	0.269	0.596	0.326	0.461	0.340	0.475
Observations	3,942	3,323	3,925	7,180	7,705	1,484	4,418	3,576	3,038	3,859

Table 4 Circumstance variables used

Countries	Circumstance variables included	Number of circumstance groups
Austria, Ireland, Luxembourg, the Netherlands, Slovenia	Sex, age, parental education, occupation father, immigration status	32
Belgium	Sex, age, parental education, occupation father, presence parents growing up	32
Bulgaria, Hungary, United Kingdom	Sex, age, parental education, occupation father	16
Germany	Sex, age, parental education, occupation father, presence parents growing up, immigration status	64

Next, we determine which of these circumstance variables are relevant for each country and should be included in the analysis.⁴⁰ To do so, we regress the set of circumstance variables on gross income for each country. Variables without a statistically significant effect are excluded from the set of circumstance variables for that country, reducing the risk of overfitting. The resulting sets of circumstance variables are presented in Table 4.⁴¹

We turn to our selection of responsibility variables. Following previous studies that have applied the liberal approach to inequality of opportunity measurement, such as Almås et al. (2011), Devooght (2008) and Ramos and Van de gaer (2021), we use marital status (married or not), hours worked (less than 20 hours, 20–30 hours, or more than 30 hours), and own education (upper secondary education and lower education levels, post-secondary non-tertiary education, or first and second stage of tertiary education).⁴² We end up with 18 responsibility groups for each country.⁴³

6.2 Results

We present results for both the absolute and the relative case. This stands in contrast to previous empirical work on inequality of opportunity, which has focused virtually exclusively on the relative case. There is good reason to broaden the analysis in this way. As the literature on income inequality measurement stresses, the choice between the absolute and relative cases

⁴⁰ Our method for selecting circumstances groups is inspired by Brunori et al. (2019), though our method is simpler and less refined. Their method is tailored to a parametric ex ante utilitarian approach, making it unsuitable for our purpose. However, like Brunori et al. (2019), our method results in different sets of circumstance groups across countries, a feature for which they provide a strong argumentation (p. 645).

⁴¹ There are still relatively many circumstance groups with few members. For Austria there are 7 circumstance groups with less than 25 members, for Belgium there are 13, for Bulgaria 4, for Germany 31, for Hungary 1, for Ireland 14, for Luxembourg 2, for the Netherlands 15, for Slovenia 12 and for the United Kingdom 0.

⁴² Responsibility variables tend to be more contentious than circumstance variables. From the control view on responsibility, which identifies responsibility variables with factors under individual control, all three of the variables we use can be debated. See Roemer and Trannoy (2016, pp. 1310–1312) for a discussion. Note that the Roemerian approach, which measures responsibility by an individual's income rank within his circumstance group, has its own challenges (Ramos and Van de gaer 2016, pp. 872–873).

⁴³ For Austria there are 4 responsibility groups with less than 25 members, for Belgium there are 2, for Bulgaria 8, for Germany 3, for Hungary 6, for Ireland 2, for Luxembourg 4, for the Netherlands 1, for Slovenia 6 and for the United Kingdom 2.

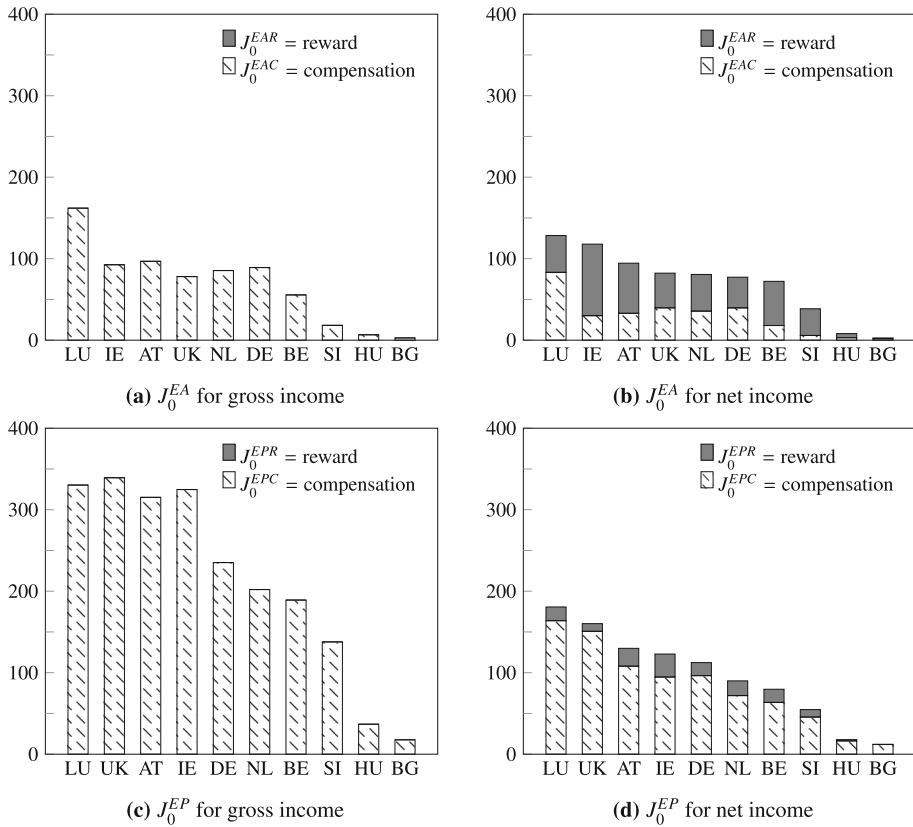


Fig. 1 J_0^{EA} and J_0^{EP} for the gross and net income distributions (in millions)

is not a positive matter: there is no right or wrong. Rather, absolute and relative measures of income inequality and divergence capture different perspectives on how to make inequality comparisons as total income changes. And although the absolute case comes with a practical disadvantage—as incomes have to be converted so that they are expressed in a comparable unit—this (easily overcome) disadvantage is not an excuse for altogether dropping what is a perfectly valid perspective on inequality.⁴⁴ Moreover, in the literature on equality of opportunity, the absolute version of liberal reward has received far more attention than the relative version, which provides a strong reason not to neglect the absolute case when empirically applying liberal measures of inequality of opportunity.

⁴⁴ On these points, see, e.g., Kolm (1976), Moyes (1987), Ravallion (2003) and Subramanian (2015). The absolute case receives substantial attention in the study of the evolution of global income inequality (e.g., Atkinson and Brandolini 2010, Bosmans et al. 2014 and Niño-Zarazúa et al. 2017), where it is shown to lead to dramatically different conclusions compared to the relative case. Questionnaire results support extending the analysis beyond the relative case. For example, Amiel and Cowell (1999a, p. 38) find that while the relative case receives most support (about a third of the respondents), the absolute case comes second (supported by about a sixth) and intermediate positions receive the bulk of the remaining support. See also Amiel and Cowell (1992, 1999b), Ballano and Ruiz-Castillo (1993) and Harrison and Seidl (1994).

Figure 1 presents the values of the absolute inequality of opportunity measures as well as their compensation and reward components (see also Table 5).⁴⁵ Panels (a) and (b) provide the results for the ex ante measure for gross income (i.e., x_N is set equal to the gross income distribution in Eq. 8) and for net income (i.e., x_N is set equal to the net income distribution in Eq. 8), respectively. Panels (c) and (d) do the same for the ex post measure in Eq. 11. The measures are computed for $\alpha = 0$, meaning that the income inequality measure used to compute the compensation and reward components is the variance. Countries are ranked on the basis of overall inequality of opportunity in net income.⁴⁶

We first focus on the results for the net income distribution in panels (b) and (d). With the exception of Austria, Germany, Ireland and the Netherlands, the ranking of the countries by the ex ante and ex post measures in terms of overall inequality of opportunity is similar.⁴⁷ However, in terms of the compensation and reward components, the ex ante and ex post measures paint a very different picture. For the ex ante measure, the reward component is substantial, and with the exception of Bulgaria, Germany and Luxembourg, larger than the compensation component. For the ex post measure, in contrast, the compensation component dominates.⁴⁸

Our results are in line with the common finding in the literature that the values of the ex ante utilitarian measure (our ex ante compensation component) are smaller than those of the ex post utilitarian measure (our ex post compensation component).⁴⁹

Note that, by construction, the value of the inequality of opportunity measure is greater in the liberal approach than in the utilitarian approach. Our decomposition thus helps explain the observation that the inequality of opportunity to income inequality ratios found in the liberal approach are greater than those in the utilitarian approach.⁵⁰ For the ten countries under consideration, we find that the (liberal) average ratio of overall inequality of opportunity to income inequality is 0.53 for the ex ante measure and 0.79 for the ex post measure,

⁴⁵ To assess the precision of the point estimates in Table 5, we used the jackknife method. This method involves excluding one observation at a time, recomputing the relevant measure, and estimating confidence intervals. The 95% confidence intervals around the point estimates are narrow: across all countries and all inequality of opportunity measures and components, the largest margin of error is 0.02775 million for the absolute measures in the top half of Table 5 and 0.00001 for the relative measures in the bottom half.

⁴⁶ As mentioned above, all incomes are expressed in PPP euros by multiplying them by a country-specific conversion factor. Because we use the variance, the numbers in Fig. 1 and Table 5 can be converted back to the PPP-unadjusted numbers by dividing them by the square of the country's conversion factor. The squared conversion factors are 0.806 for Austria, 0.768 for Belgium, 4.594 for Bulgaria, 0.905 for Germany, 2.922 for Hungary, 0.639 for Ireland, 0.516 for Luxembourg, 0.763 for the Netherlands, 1.720 for Slovenia and 0.794 for the United Kingdom. Clearly then, the PPP adjustment affects comparisons between countries, but does not qualitatively affect comparisons of levels of inequality of opportunity and its components within each country.

⁴⁷ Although absolute income inequality or inequality of opportunity comparisons often follow mean income comparisons (since higher mean income allows for greater absolute income differences), several instances in our data deviate from this pattern. For example, the United Kingdom has lower mean net income than Ireland, but exhibits greater net income inequality. Similarly, Germany has lower mean gross and net incomes than the Netherlands, but shows greater ex post inequality of opportunity for both net and gross income.

⁴⁸ On average, the share of the reward component in overall inequality of opportunity is 58.8 percent for the ex ante measure and 13.6 percent for the ex post measure.

⁴⁹ See, for example, Checchi et al. (2016) and Ramos and Van de gaer (2016).

⁵⁰ This observation is made, for example, by Ramos and Van de gaer (2016, p. 877). Our analysis supports their suspicion that the observation is systematic rather than coincidental.

whereas the (utilitarian) average ratio of the compensation component to income inequality is, respectively, 0.20 and 0.68.⁵¹

Next, we turn to the gross income distribution in panels (a) and (c). We first note that, by construction, the reward component is zero for both the ex ante and ex post measures.⁵² This reflects that the gross income distribution, which we have equated with the market income distribution, is ideal in terms of reward.

A comparison of the inequality of opportunity values for the net and gross income distributions reveals how inequality of opportunity is affected by the tax system. An exclusive focus on the compensation component, as in the utilitarian approach, yields a clear-cut conclusion: the tax system of each of the ten countries reduces inequality of opportunity. However, the conclusion is no longer clear-cut if, as in the liberal approach, we also consider the reward component. Then the impact of the tax system depends on the magnitudes of two opposed effects: a decrease in the compensation component and an increase in the reward component.

For the ex ante inequality of opportunity measure, the reward effect offsets the compensation effect for five out of ten countries (Belgium, Hungary, Ireland, Slovenia and the United Kingdom), yielding the surprising conclusion that the tax system increases inequality of opportunity. Consider, for example, the United Kingdom in Table 5. When moving from gross to net income, the value of the ex ante measure increases from 78.02 to 82.25. This increase is the net result of a decrease in the compensation component from 78.02 to 39.66 and an increase in the reward component from 0 to 42.59. In contrast, for the ex post inequality of opportunity measure, the compensation effect always outweighs the reward effect.

We now turn to the relative case. Figure 2 presents the values of the relative ex ante and ex post inequality of opportunity measures and their components for net and gross income (see also Table 5). We set $\rho = 0$, which means that the income inequality measure used for the compensation components is the mean logarithmic deviation.

A striking difference with the absolute case is that the reward component for net income is considerably less pronounced for both the ex ante (panel (b)) and ex post (panel (d)) measures. A plausible explanation of this difference is that tax systems are closer to preserving relative rather than absolute income differences, causing them to perform better in terms of relative than absolute versions of reward. A word of caution is in order. Echoing our discussion at the start of this subsection, these findings should not be interpreted as saying that relative versions of reward are more appropriate than their absolute counterparts. Rather, the absolute and relative versions of reward reflect different ethical values and the choice between them should not depend on the observed properties of existing tax systems.⁵³ Whether Fig. 1 or Fig. 2 holds the relevant results ultimately depends on whether one subscribes to the absolute or relative version of reward.

Another difference with the absolute case is that the rankings of the countries in terms of overall inequality of opportunity by the ex ante and ex post measures are very different both for the net and the gross income distributions. However, as in the absolute case, for both net

⁵¹ It is common in the literature to report the ratio of inequality of opportunity to income inequality. This ratio is meaningful, as it indicates what proportion of income inequality can be regarded as ethically problematic. However, it would be wrong to regard this ratio as a mere convenient normalization of inequality of opportunity. While the ratio may be easier to interpret in a cardinal sense, it does not preserve the ordinal inequality of opportunity rankings. For example, in our results (see Table 5), Austria exhibits greater inequality of opportunity than Belgium across all measures (ex ante and ex post, absolute and relative), but has a lower ratio of inequality of opportunity to income inequality.

⁵² To see this, let $x_N = m_N$ in Eqs. 10 and 13.

⁵³ This choice is closely related to that between equal absolute and equal proportional taxes for the role of the inequality-neutral benchmark in the measurement of tax progressivity. See, e.g., Urban (2019).

Table 5 J_0^{EA} (in millions), J_0^{EP} (in millions), J_0^{REA} and J_0^{REP} for the net and gross income distributions

Country	J_0^{EA}		J_0^{EAC}		J_0^{EAR}		J_0^{EP}		J_0^{EPC}		J_0^{EPR}		$I_{\alpha=0}$	
	Net	Gross	Net	Gross	Net	Gross	Net	Gross	Net	Gross	Net	Gross	Net	Gross
AT	94.47	96.81	32.98	96.81	61.49	0	129.91	315.17	107.97	315.17	21.93	0	151.74	442.49
BE	72.21	55.53	17.98	55.53	54.23	0	79.68	189.14	63.54	189.14	16.13	0	86.50	265.88
BG	2.71	2.75	1.81	2.75	0.91	0	12.08	17.62	11.94	17.62	0.14	0	13.63	20.41
DE	77.27	89.12	39.62	89.12	37.65	0	112.37	235.12	96.37	235.12	16.00	0	169.32	391.31
HU	8.09	6.61	2.96	6.61	5.13	0	17.83	36.82	16.29	36.82	1.55	0	23.15	51.69
IE	117.85	92.55	30.07	92.55	87.78	0	122.87	324.74	94.67	324.74	28.21	0	150.81	486.15
LU	128.41	162.04	83.22	162.04	45.19	0	180.67	330.26	163.79	330.26	16.87	0	253.16	509.98
NL	80.65	85.43	35.66	85.43	44.99	0	89.94	202.02	71.87	202.02	18.07	0	142.54	358.14
SI	38.56	18.23	5.78	18.23	32.78	0	54.73	137.78	45.56	137.78	9.17	0	61.66	187.27
UK	82.25	78.02	39.66	78.02	42.59	0	160.18	339.23	150.76	339.23	9.42	0	220.58	469.22
Country	J_0^{REA}		J_0^{REAC}		J_0^{REAR}		J_0^{REP}		J_0^{REPC}		J_0^{REPR}		$I_{\rho=0}$	
	Net	Gross	Net	Gross	Net	Gross	Net	Gross	Net	Gross	Net	Gross	Net	Gross
AT	0.0386	0.0478	0.0347	0.0478	0.0039	0	0.1280	0.1595	0.1264	0.1595	0.0016	0	0.1903	0.2466
BE	0.0309	0.0267	0.0204	0.0267	0.0105	0	0.0754	0.0928	0.0740	0.0928	0.0014	0	0.1069	0.1396
BG	0.0253	0.0228	0.0227	0.0228	0.0026	0	0.1344	0.1328	0.1343	0.1328	0.0001	0	0.1581	0.1589
DE	0.0373	0.0457	0.0340	0.0457	0.0033	0	0.0894	0.1194	0.0879	0.1194	0.0015	0	0.1920	0.2511
HU	0.0183	0.0214	0.0165	0.0214	0.0018	0	0.1021	0.1216	0.1014	0.1216	0.0007	0	0.1467	0.1755
IE	0.0408	0.0445	0.0299	0.0445	0.0109	0	0.1016	0.1465	0.0982	0.1465	0.0034	0	0.1886	0.2713
LU	0.0631	0.0677	0.0577	0.0677	0.0054	0	0.1172	0.1333	0.1161	0.1333	0.0011	0	0.1931	0.2220
NL	0.0275	0.0349	0.0216	0.0349	0.0059	0	0.0598	0.0852	0.0569	0.0852	0.0029	0	0.1201	0.1715
SI	0.0161	0.0173	0.0119	0.0173	0.0042	0	0.1317	0.1570	0.1304	0.1570	0.0013	0	0.1625	0.2013
UK	0.0429	0.0475	0.0414	0.0475	0.0015	0	0.1254	0.1459	0.1251	0.1459	0.0003	0	0.2345	0.2604

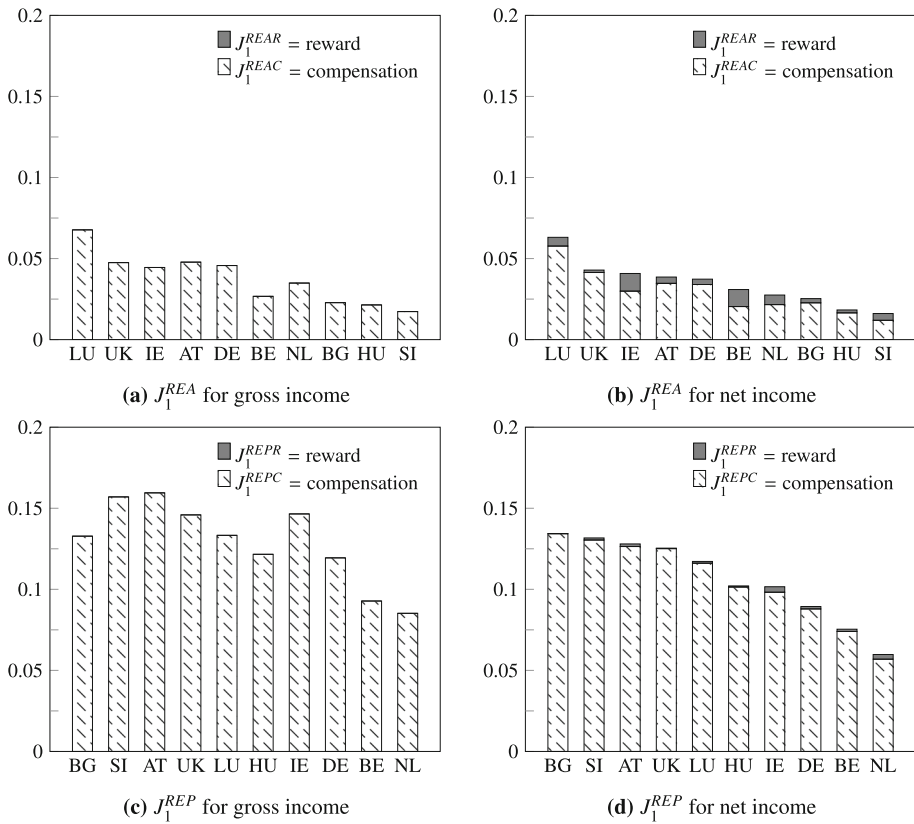


Fig. 2 J_I^{REA} and J_I^{REP} for the gross and net income distributions

and gross income, the compensation component (which in the utilitarian approach represents overall inequality of opportunity) is smaller for the ex ante case than for the ex post case. Moreover, again, for net income, the reward component of the ex ante measure is larger than that of the ex post measure.⁵⁴ Note that, in the relative case, the tax system reduces inequality of opportunity in both the ex ante and the ex post cases for all countries, except Belgium in the ex ante case, and Belgium and Bulgaria in the ex post case.

Our empirical results yield several interesting conclusions. First, our decomposition, especially for the ex ante case, reveals that a sizeable part of absolute inequality of opportunity can be attributed to reward. This result is noteworthy since previous research using the utilitarian approach has exclusively focused on the compensation component, thus arriving at considerably lower levels of inequality of opportunity expressed as a percentage of income inequality. In the relative case, the reward component is far less important. An explanation could be that tax systems are better at preserving relative market income differentials than absolute ones. Second, although taxation in our data almost always decreases the compensation component of inequality of opportunity,⁵⁵ it by construction always increases the reward component. Thus, whether or not taxation increases or decreases total inequality of opportunity depends

⁵⁴ On average, the share of the reward component in overall inequality of opportunity is 15.9 percent for the ex ante measure and 1.6 percent for the ex post measure.

⁵⁵ The only exception is Bulgaria in the relative ex post case.

on the relative magnitudes of the decrease of the compensation component and the increase of the reward component. This potentially leads to strongly different evaluations of tax systems between, on the one hand, the utilitarian approach that considers compensation only and, on the other hand, the liberal approach that in addition considers reward.

7 Conclusion

We have developed ex ante and ex post measures of inequality of opportunity that decompose neatly into compensation and reward parts. The compensation part measures the inequality between individuals with different circumstances. The reward part measures the extent to which income differences reflect market returns to the exercise of responsibility. An important observation is that the compensation parts of both measures take the form of utilitarian measures, thus revealing a striking connection between the rival liberal and utilitarian approaches to the measurement of inequality of opportunity.

We have applied the newly developed measures and their decompositions to the net and gross incomes distributions of ten European countries. The decompositions prove to be helpful tools to further insight. For example, using the absolute ex ante measure, we find that the tax systems of half of the countries under consideration increase inequality of opportunity, and the decomposition reveals that this is the result of a decrease in compensation unfairness that is offset by a greater increase in reward unfairness. Our empirical analysis showcases, moreover, the added value of being able to study liberal and utilitarian approaches to inequality of opportunity within a single unified framework.

For future research, it would be valuable to study combinations of alternative fair division rules, such as the egalitarian equivalent and conditionally egalitarian rules (Bossert and Fleurbaey 1996), and alternative divergence measures. Examining the decomposability properties of the resulting inequality of opportunity measures could prove insightful. At a more fundamental level, it would be worthwhile to axiomatically characterize inequality of opportunity measures directly, without relying on the two-step approach that combines fair division rules and divergence measures. This approach could include decomposability into compensation and reward parts as an explicit axiom. It would be interesting to see how the inequality of opportunity measures that emerge from such an axiomatic analysis compare with those we have studied here.

Acknowledgements We thank audiences in Bari (Equal Chances: Equality of Opportunity and Social Mobility Around the World Conference), Maastricht (Dutch Social Choice Colloquium), Paris (Eighth Meeting of the Society for the Study of Economic Inequality), Tokyo (Hokkaido Summer Institute 2019 - 5th Hitotsubashi Institute), Tilburg (Economics Workshop, Tilburg University) and Portsmouth (Economics and Finance Subject Group Research Seminar, University of Portsmouth) for useful comments. This study is based on data from Eurostat, EU Statistics on Income and Living Conditions 2011. The responsibility for all conclusions drawn from the data lies entirely with the authors.

Author Contributions All three authors have contributed equally to all parts of the manuscript.

Funding Not applicable.

Data Availability Researchers may apply to Eurostat for access to the data for scientific purposes.

Declarations

Ethical Approval Not applicable.

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Alesina, A., La Ferrara, E.: Preferences for redistribution in the land of opportunities. *J. Public Econ.* **89**(5–6), 897–931 (2005)
- Almås, I., Cappelen, A.W., Lind, J.T., Sørensen, E.Ø., Tungodden, B.: Measuring unfair (in)equality. *J. Public Econ.* **95**(7), 488–499 (2011)
- Almås, I., Cappelen, A.W., Tungodden, B.: Cutthroat capitalism versus cuddly socialism: Are Americans more meritocratic and efficiency-seeking than Scandinavians? *J. Polit. Econ.* **128**(5), 1753–1788 (2020)
- Álvarez, A.S., Menéndez, A.J.L.: Assessing changes over time in inequality of opportunity: the case of Spain. *Soc. Indic. Res.* **139**(3), 989–1014 (2018a)
- Álvarez, A.S., Menéndez, A.J.L.: Income inequality and inequality of opportunity in Europe: are they on the rise? In: Bishop, J.A., Rodríguez, J.G. (eds.) *Inequality, Taxation and Intergenerational Transmission* (Research on Economic Inequality, Volume 26), pp. 149–196. Emerald Group Publishing Limited (2018b)
- Amiel, Y., Cowell, F.A.: Measurement of income inequality: Experimental test by questionnaire. *J. Public Econ.* **47**(1), 3–26 (1992)
- Amiel, Y., Cowell, F.A.: Income transformations and income inequality. In: Slottje, D. (ed.) *Advances in Econometrics, Income Distribution and Scientific Methodology*, pp. 209–232. Physica Verlag, Heidelberg (1999a)
- Amiel, Y., Cowell, F.A.: *Thinking about Inequality*. Cambridge University Press, Cambridge (1999b)
- Arneson, R.J.: Equality and equal opportunity for welfare. *Philos. Stud.* **56**(1), 77–93 (1989)
- Atkinson, A.B.: On the measurement of inequality. *J. Econ. Theory* **2**(3), 244–263 (1970)
- Atkinson, A.B., Brandolini, A.: On analyzing the world distribution of income. *World Bank Econ. Rev.* **24**(1), 1–37 (2010)
- Ballano, C., Ruiz-Castillo, J.: Searching by questionnaire for the meaning of income inequality. *Revista española de economía* **10**, 233–259 (1993)
- Belhaj Hassine, N.: Inequality of opportunity in Egypt. *World Bank Econ. Rev.* **26**(2), 265–295 (2012)
- Bosmans, K., Cowell, F.A.: The class of absolute decomposable inequality measures. *Econ. Lett.* **109**(3), 154–156 (2010)
- Bosmans, K., Decancq, K., Decoster, A.: The relativity of decreasing inequality between countries. *Economica* **81**, 276–292 (2014)
- Bosmans, K., Öztürk, Z.E.: Measurement of inequality of opportunity: A normative approach. *J. Econ. Inequal.* **19**(2), 213–237 (2021)
- Bosmans, K., Öztürk, Z.E.: Equality and responsibility: ex ante and ex post redistribution mechanisms. *Econ. Theor.* **73**(4), 947–975 (2022)
- Bossert, W.: Redistribution mechanisms based on individual characteristics. *Math. Soc. Sci.* **29**(1), 1–17 (1995)
- Bossert, W., Fleurbaey, M.: Redistribution and compensation. *Soc. Choice Welfare* **13**(3), 343–355 (1996)
- Bossert, W., Fleurbaey, M., Van de gaer, D.: Responsibility, talent, and compensation: a second-best analysis. *Rev. Econ. Design* **4**(1), 35–55 (1999)
- Bourguignon, F.: Decomposable income inequality measures. *Econometrica* **47**(4), 901–920 (1979)
- Bourguignon, F., Ferreira, F.H.G., Menéndez, M.: Inequality of opportunity in Brazil. *Rev. Income Wealth* **53**(4), 585–618 (2007)
- Brunori, P., Palmisano, F., Peragine, V.: Inequality of opportunity in sub-Saharan Africa. *Appl. Econ.* **51**(60), 6428–6458 (2019)
- Brunori, P., Peragine, V., Serlenga, L.: Upward and downward bias when measuring inequality of opportunity. *Soc. Choice Welfare* **52**(4), 635–661 (2019)

- Cappelen, A.W., Hole, A.D., Sørensen, E.Ø., Tungodden, B.: The pluralism of fairness ideals: An experimental approach. *Am. Econ. Rev.* **97**(3), 818–827 (2007)
- Cappelen, A.W., Konow, J., Sørensen, E.Ø., Tungodden, B.: Just luck: An experimental study of risk-taking and fairness. *Am. Econ. Rev.* **103**(4), 1398–1413 (2013)
- Cappelen, A.W., Tungodden, B.: Rewarding effort. *Econ. Theor.* **39**(3), 425–441 (2009)
- Cappelen, A.W., Tungodden, B.: Fairness and the proportionality principle. *Soc. Choice Welfare* **49**(3), 709–719 (2017)
- Chakravarty, S.R.: The variance as a subgroup decomposable measure of inequality. *Soc. Indic. Res.* **53**(1), 79–95 (2001)
- Chakravarty S.R., Tyagarupananda, S.: The subgroup decomposable absolute indices of inequality. In: Chakravarty, S.R., Coondoo, D., Mukherjee, R. (eds) *Quantitative Economics: Theory and Practice, Essays in Honor of Professor N. Bhattacharya*, pp. 247–257. New Delhi: Allied Publishers (1998)
- Chakravarty, S.R., Tyagarupananda, S.: The subgroup decomposable intermediate indices of inequality. *Span Econ Rev* **11**(2), 83–97 (2009)
- Checchi, D., Peragine, V.: Inequality of opportunity in Italy. *J. Econ. Inequal.* **8**(4), 429–450 (2010)
- Checchi, D., Peragine, V., Serlenga, L.: Inequality of opportunity in Europe: Is there a role for institutions? In: Cappellari, L., Polachek, S.W., Tatsiramos, K. (eds) *Inequality: Causes and Consequences (Research in Labor Economics, vol. 43)*, pp. 1–44. Bingley: Emerald Publishing Ltd (2016)
- Cohen, G.A.: On the currency of egalitarian justice. *Ethics* **99**(4), 906–944 (1989)
- Cowell, F.A.: Generalized entropy and the measurement of distributional change. *Eur. Econ. Rev.* **13**, 147–159 (1980a)
- Cowell, F.A.: On the structure of additive inequality measures. *Rev. Econ. Stud.* **47**(3), 521–531 (1980b)
- Cowell, F.A.: Measures of distributional change: an axiomatic approach. *Rev. Econ. Stud.* **52**(1), 135–151 (1985)
- Cowell, F.A., Kuga, K.: Additivity and the entropy concept: an axiomatic approach to inequality measurement. *J. Econ. Theory* **25**(1), 131–143 (1981a)
- Cowell, F.A., Kuga, K.: Inequality measurement: an axiomatic approach. *Eur. Econ. Rev.* **15**(3), 287–305 (1981b)
- Devooght, K.: To each the same and to each his own: a proposal to measure responsibility-sensitive income inequality. *Economica* **75**(298), 280–295 (2008)
- Dworkin, R.: What is equality? Part 1: Equality of welfare. *Philos. Public Aff.* **10**(4), 185–246 (1981a)
- Dworkin, R.: What is equality? Part 2: Equality of resources. *Philos. Public Aff.* **10**(4), 283–345 (1981b)
- Ferreira, F.H.G., Gignoux, J.: The measurement of inequality of opportunity: Theory and an application to Latin America. *Rev. Income Wealth* **57**(4), 622–657 (2011)
- Ferreira, F.H.G., Peragine, V.: Individual responsibility and equality of opportunity. In: Adler, M.D., Fleurbaey, M. (eds.) *The Oxford Handbook of Well-Being and Public Policy*, pp. 746–784. Oxford University Press, New York (2016)
- Fleurbaey, M.: On fair compensation. *Theor. Decis.* **36**(3), 277–307 (1994)
- Fleurbaey, M.: Three solutions for the compensation problem. *J. Econ. Theory* **65**(2), 505–521 (1995)
- Fleurbaey, M.: Fairness, Responsibility, and Welfare. Oxford University Press, New York (2008)
- Fleurbaey, M., Maniquet, F.: Compensation and responsibility. In: Arrow, K.J., Sen, A., Suzumura, K. (eds.) *Handbook of Social Choice and Welfare*, vol. II, pp. 507–604. North-Holland, Amsterdam (2011)
- Fleurbaey, M., Peragine, V.: Ex ante versus ex post equality of opportunity. *Economica* **80**(317), 118–130 (2013)
- Fleurbaey, M., Schokkaert, E.: Unfair inequalities in health and health care. *J. Health Econ.* **28**(1), 73–90 (2009)
- Fong, C.: Social preferences, self-interest, and the demand for redistribution. *J. Public Econ.* **82**(2), 225–246 (2001)
- Harrison, E., Seidl, C.: Perceptual inequality and preferential judgements: An empirical examination of distributional axioms. *Public Choice* **79**, 61–81 (1994)
- Hufe, P., Kanbur, R., Peichl, A.: Measuring unfair inequality: reconciling equality of opportunity and freedom from poverty. *Rev. Econ. Stud.* **89**(6), 3345–3380 (2022)
- Hufe, P., Peichl, A., Roemer, J., Ungerer, M.: Inequality of income acquisition: the role of childhood circumstances. *Soc. Choice Welfare* **49**(3–4), 499–544 (2017)
- Iturbe-Ormaetxe, I.: Redistribution and individual characteristics. *Rev. Econ. Design* **3**(1), 45–55 (1997)
- Kolm, S.-C.: Unequal inequalities I. *J. Econ. Theory* **12**(3), 416–442 (1976)
- Konow, J.: Fair shares: Accountability and cognitive dissonance in allocation decisions. *Am. Econ. Rev.* **90**(4), 1072–1091 (2000)
- Lefranc, A., Pistoiesi, N., Trannoy, A.: Inequality of opportunities vs. inequality of outcomes: are western societies all alike? *Rev. Income Wealth* **54**(4), 513–546 (2008)

- Lefranc, A., Pistolesi, N., Trannoy, A.: Equality of opportunity and luck: definitions and testable conditions, with an application to income in France. *J. Public Econ.* **93**(11), 1189–1207 (2009)
- Mack, E., Gaus, G.F.: Classical liberalism and libertarianism: The liberty tradition. In: Gaus, G.F., Kukathas, C. (eds.) *Handbook of Political Theory*. Sage, London (2004)
- Magdalou, B., Nock, R.: Income distributions and decomposable divergence measures. *J. Econ. Theory* **146**(6), 2440–2454 (2011)
- Mahler, D.G., Ramos, X.: Equality of opportunity in four measures of well-being. *Rev. Income Wealth* **65**(S1), S228–S255 (2019)
- Marrero, G.A., Rodríguez, J.G.: Inequality of opportunity in Europe. *Rev. Income Wealth* **58**(4), 597–621 (2012)
- Marrero, G.A., Rodríguez, J.G.: Inequality of opportunity and growth. *J. Dev. Econ.* **104**, 107–122 (2013)
- Moyes, P.: A new concept of Lorenz domination. *Econ. Lett.* **23**(2), 203–207 (1987)
- Niño-Zarazúa, M., Roope, L., Tarp, F.: Global inequality: relatively lower, absolutely higher. *Rev. Income Wealth* **63**(4), 661–684 (2017)
- Ooghe, E., Schokkaert, E., Van de gaer, D.: Equality of opportunity versus equality of opportunity sets. *Soc. Choice Welfare* **28**(2), 209–230 (2007)
- Ramos, X., Van de gaer, D.: Approaches to inequality of opportunity: principles, measures and evidence. *J. Econ. Surveys* **30**(5), 855–883 (2016)
- Ramos, X., Van de gaer, D.: Is inequality of opportunity robust to the measurement approach? *Rev. Income Wealth* **67**(1), 18–36 (2021)
- Ravallion, M.: The debate on globalization, poverty and inequality: why measurement matters. *Int. Aff.* **79**(4), 739–753 (2003)
- Rawls, J.: *A Theory of Justice*. Belknap Press of Harvard University Press, Cambridge, Mass (1971)
- Rawls, J.: Social unity and primary goods. In: Sen, A.K., Williams, B. (eds.) *Utilitarianism and Beyond*. Cambridge University Press, Cambridge (1982)
- Roemer, J.E.: A pragmatic theory of responsibility for the egalitarian planner. *Phil. Pub. Aff.* **22**(2), 146–166 (1993)
- Roemer, J.E.: On several approaches to equality of opportunity. *Econ. Philos.* **28**(2), 165–200 (2012)
- Roemer, J.E., Trannoy, A.: Equality of opportunity: theory and measurement. *J. Econ. Lit.* **54**(4), 1288–1332 (2016)
- Schokkaert, E., Devooght, K.: Responsibility-sensitive fair compensation in different cultures. *Soc. Choice Welfare* **21**(2), 207–242 (2003)
- Shorrocks, A.F.: The class of additively decomposable inequality measures. *Econometrica* **48**(3), 613–625 (1980)
- Shorrocks, A.F.: Inequality decomposition by population subgroups. *Econometrica* **52**(6), 1369–1385 (1984)
- Subramanian, S.: Assessing inequality in the presence of growth: an expository essay. *Rev. Black Polit. Econ.* **42**(3), 179–199 (2015)
- Trannoy, A.: Are the theories of equality of opportunity put forward by Fleurbaey and Roemer incompatible? *Revue économique* **68**(1), 57–72 (2017)
- Urban, I.: Measuring redistributive effects of taxes and benefits: beyond the proportionality standard. *FinanzArchiv* **75**(4), 413–443 (2019)
- Van de gaer, D.: *Equality of Opportunity and Investment in Human Capital*. Katholieke Universiteit Leuven, Faculteit der Economische en Toegepaste Economische Wetenschappen (1993)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.